

PakPulse-AI

Multi-Disease Outbreak Forecasting & Early Warning System for Pakistan

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Table Of Content

1	INTRODUCTION	1
1.1	SYSTEM INTRODUCTION	1
1.2	BACKGROUND OF THE SYSTEM.....	1
1.3	OBJECTIVES OF THE SYSTEM	1
1.4	SIGNIFICANCE OF THE SYSTEM	2
2	OVERALL DESCRIPTION.....	3
2.1	PRODUCT PERSPECTIVE	3
2.2	PRODUCT SCOPE.....	3
2.3	PRODUCT FUNCTIONALITY	4
2.4	USERS AND CHARACTERISTICS	5
	ADMINISTRATOR	5
	HEALTH OFFICERS.....	6
2.5	OPERATING ENVIRONMENT	6
3	SPECIFIC REQUIREMENTS.....	8
3.1	FUNCTIONAL REQUIREMENTS.....	8
	<i>Data Collection & Storage Module:</i>	8
3.2	BEHAVIOUR REQUIREMENTS	9
3.3	EXTERNAL INTERFACE REQUIREMENTS.....	11
3.4	USER INTERFACES	11
3.4.1	<i>Hardware Interfaces</i>	12
	<i>No hardware interfaces will be used in our system.</i>	12
3.4.2	<i>Software Interfaces</i>	13
3.4.3	<i>Communications Interfaces</i>	14
4	OTHER NON-FUNCTIONAL REQUIREMENTS.....	15
4.1	PERFORMANCE REQUIREMENTS	15
4.1.1	<i>Response Time</i>	15
4.1.2	<i>Availability</i>	15
4.1.3	<i>Accuracy</i>	15
4.2	SAFETY AND SECURITY REQUIREMENTS	15
4.3	SOFTWARE QUALITY ATTRIBUTES	15
4.3.1	<i>Usability</i>	15
4.3.2	<i>User Interface</i>	15
4.3.3	<i>Ease of Use:</i>	16
	<i>Users should be able to navigate the system and access required features within 5 minutes of first use</i>	16
4.3.4	<i>Learnability</i>	16
4.3.5	<i>Reliability</i>	16
4.3.6	<i>Availability</i>	16
4.3.7	<i>Robustness</i>	16
4.3.8	<i>Portability</i>	16
5	DESIGN DESCRIPTION	17
5.1	COMPOSITE VIEWPOINT.....	17
5.2	LOGICAL VIEWPOINT	18
5.3	INFORMATION VIEWPOINT.....	19
5.4	INTERACTION VIEWPOINT.....	20
5.4.1	<i>Sequence Diagram for General User</i>	20
5.4.2	<i>Sequence Diagram for Admin</i>	21
5.4.3	<i>Sequence Diagram for Health Officer</i>	22

5.5	STATE DYNAMICS VIEWPOINT.....	23
5.5.1	<i>User State Diagram</i>	23
5.5.2	<i>Admin State Diagram</i>	24
5.5.3	<i>Health Officer State Diagram</i>	25
5.6	ALGORITHM VIEWPOINT.....	26
5.6.1	<i>Algorithmic Viewpoint — Login (General System Login):</i>	26
5.6.2	<i>Algorithmic Viewpoint — Admin</i>	27
5.6.3	<i>Algorithmic Viewpoint — Health Officer:</i>	28

Table Of Figures

Figure 3.1 Usecase Diagram of PakPulse-AI	10
Figure 3.2 GIS Visuals.....	11
Figure 3.3 Dashboard.....	12
Figure 5.1 Composite view of PakPulse-AI	17
Figure 5.2 Class Diagram	18
Figure 5.3 ER Diagram PakPulse-AI.....	19
Figure 5.4 Sequence Diagram General User.....	20
Figure 5.5 Sequence Diagram Admin.....	21
Figure 5.6 Sequence Diagram-Health Officer	22
Figure 5.7 User State Diagram.....	23
Figure 5.8 State Diagram Health Officer.....	23
Figure 5.9 State Diagram Admin	24
Figure 5.10 State Diagram Health Officer.....	25

1 Introduction

1.1 System Introduction

PakPulse AI is an integrated, AI-powered disease surveillance and forecasting system designed to support Pakistan's public health sector. The system provides a unified platform where health officials can monitor multiple infectious diseases, view district-level outbreak risks, analyze forecast trends, and receive timely alerts. By combining machine learning, deep learning, epidemiological indicators, and GIS-based visualization, the system enables early detection of potential outbreaks. The platform shifts the conventional reactive approach toward a proactive early-warning model, helping authorities prepare in advance and respond more effectively.

1.2 Background of the System

Pakistan frequently faces seasonal disease outbreaks—such as dengue, malaria, cholera, influenza, and COVID-19—which strain public health infrastructure. Although various dashboards and reporting channels exist, they are typically disease-specific, dependent on late hospital reporting, and lack predictive or explainable capabilities. There is no unified national framework that forecasts multiple diseases together or translates data into actionable insights.

PakPulse AI addresses this gap by integrating real-time weather data, global health datasets, and simulated hospital reports into a centralized system. With machine learning-based detection, deep learning-based forecasting, and explainable AI methods, the platform introduces a modern yet practical way to foresee upcoming health risks while still aligning with established public health procedures.

1.3 Objectives of the System

The main objective of PakPulse AI is to strengthen Pakistan's disease surveillance landscape by providing reliable, data-driven early warnings. The system is designed to achieve the following:

- 1. **Predict outbreaks before they occur** through ML classifiers and deep learning models forecasting 2–4 weeks ahead.*
- 2. **Provide a unified, multi-disease monitoring platform** instead of relying on separate*

dashboards for each disease algorithm.

3. **Generate district-wise Composite Epidemic Risk Scores (CERS)** to simplify complex data into a clear risk indicator.
4. **Present insights visually** through GIS-based maps, interactive charts, and risk meters.
5. **Enable timely action** using automated alerts, weekly reports, and rule-based recommendations for health interventions

1.4 Significance of the System

PakPulse AI brings several important advantages to Pakistan's public health sector:

• Early Preparedness

By forecasting likely outbreaks weeks in advance, health departments can plan fumigation campaigns, medicine stockpiling, bed allocation, and awareness drives ahead of time.

• Unified Disease Management

Instead of monitoring dengue, malaria, influenza, and other diseases separately, PakPulse AI provides a consolidated multi-disease platform, reducing fragmentation and improving coordination.

• Precision and Reliability

Through machine learning and climate-health correlations, the system improves the accuracy of outbreak prediction, reducing dependency on delayed hospital reporting.

• Improved Decision-Making

The Composite Epidemic Risk Score (CERS) turns complex epidemiological and environmental data into simple, actionable insights that policymakers can interpret quickly.

• Enhanced Public Health Response

GIS maps, forecast graphs, trend analysis, and explainable AI help authorities not only identify risks but also understand the underlying causes, supporting smarter interventions.

• Cost-Effectiveness

Proactive planning prevents resource wastage, reduces emergency load on hospitals, and minimizes large-scale outbreak damages—saving both public funds and human lives.

• Technological Advancement

PakPulse AI introduces innovation into Pakistan's health infrastructure by applying modern AI, data science, and automated communication technologies, aligning traditional public health

2 Overall Description

2.1 Product Perspective

"PakPulse-AI" is a standalone, AI-driven outbreak forecasting and surveillance platform developed to support Pakistan's public health ecosystem. It operates as an independent software product while interacting with multiple external data sources, including WHO APIs, weather data providers, and simulated hospital reporting channels. Through a web-based dashboard, authorized health officials can monitor disease trends, assess district-level risks, and receive timely alerts. The system serves as an intelligent decision-support tool that complements traditional public health reporting mechanisms.

2.2 Product Scope

The scope of PakPulse AI includes all features and functions related to multi-disease forecasting, outbreak risk assessment, explainability, and real-time alerting. Some of the major capabilities within the product scope include:

- **Multi-Disease Forecasting**

The system predicts cases of dengue, malaria, cholera, influenza, and COVID-19 using machine learning and deep learning models, providing 2–4 week forecasts.

- **Composite Epidemic Risk Scoring (CERS)**

PakPulse AI generates district-level risk scores based on climate conditions, past outbreaks, sanitation levels, and demographic indicators, allowing policymakers to quickly interpret threat levels.

- **Explainable AI (XAI)**

The system employs SHAP and LIME to clarify why certain risks or outbreaks have been predicted, enhancing trust and transparency.

- **Integration Capabilities**

PakPulse AI connects with climate APIs, WHO datasets, and simulated hospital feeds to ensure continuous data flow. It can also integrate with health department systems for automated report distribution.

- **Interactive Visualization**

The platform offers GIS-enabled maps, forecast charts, and risk dashboards to present complex epidemiological patterns in a clear and actionable format.

- **Alerts and Notifications**

Automated email/SMS alerts and weekly reports ensure health officials are informed promptly when risks exceed predefined thresholds.

2.3 Product Functionality

The functionality of PakPulse AI covers all essential operations required for disease monitoring and forecasting. Major functions include:

- **Disease Trend Forecasting**

Machine learning and deep learning models analyze historical and current data to forecast upcoming risks for multiple diseases.

- **Risk Score Generation**

CERS provides a single, district-specific score that reflects potential outbreak severity based on environmental, health, and demographic indicators.

- **Data Collection and Management**

The system retrieves, stores, and updates datasets from WHO, weather APIs, and synthetic hospital feeds, maintaining a clean and normalized data pipeline.

- **GIS Visualization**

Interactive maps display disease intensity, forecasted trends, and high-risk zones across Pakistan.

- **Real-Time Alerting**

Email and SMS notifications are triggered when outbreak probabilities cross thresholds. Weekly PDF reports summarize key findings.

- **User Activity and Access Control**

Authentication ensures secure usage by administrators, analysts, and health officers. Role-based access determines who can view or modify system components.

- **Forecast Explanations**

XAI modules explain key drivers behind each prediction, helping officials understand climate or environmental factors influencing disease spread

2.4 Users and Characteristics

PakPulse AI interacts with three primary categories of users:

1. **Administrators**
2. **Health Officers / Analysts**
3. **General Users**

Administrator

Administrators can securely register/login into the system.

They manage user accounts, roles, and access permissions.

They oversee system operations, data pipelines, and integrations with external APIs.

Administrators can view system logs, alert history, and performance reports.

They ensure backend maintenance, including database updates and model retraining scheduling.

Health Officers

Health officers and analysts can register/login for authorized access.

They can view GIS-based outbreak maps, district-level risk scores, and forecast chart.

They receive automated alerts through email/SMS when risks exceed thresholds.

They analyze disease trends, download reports, and use insights for decision-making.

They monitor past disease activity and compare trends across regions.

General Users

General users can view limited public dashboards (if enabled by admin).

They may access summarized disease forecasts and awareness information.

They cannot modify data, access risk models, or view sensitive district-level analytics.

Their access is primarily informational, aimed at promoting public awareness and preparedness.

2.5 Operating Environment

Operating System:

Web-based environment; runs on Windows/Linux servers and accessible via modern browsers (Chrome, Firefox, Edge).

Hardware Requirements:

Cloud or local server for backend processing

GPU support (optional) for deep learning training

Standard desktops/laptops for dashboard access

Database:

PostgreSQL, storing time-series datasets and alert logs.

Programming Languages:

Python (AI/ML, backend pipeline)

Libraries and Frameworks:

TensorFlow / PyTorch (deep learning models)

Scikit-learn (ML models)

Streamlit (dashboard)

Plotly / Folium (GIS visualization)

Requests, Pandas, NumPy (data handling)

Development Environment:

VS Code for backend development

Streamlit for dashboard deployment

3 Specific Requirements

3.1 Functional Requirements

The functional requirements describe the main modules and features of PakPulse AI. These modules define the key operations that the system will perform:

1. Data Collection & Storage
2. AI & Forecasting
3. Composite Epidemic Risk Score (CERS) Generation
4. Explainable AI (XAI) Module
5. Dashboard & GIS Visualization
6. Alert & Notification Module
7. User Management & Authentication

Data Collection & Storage Module:

This module fetches health and environmental data from WHO GHO/Athena APIs, OpenWeatherMap, and simulated hospital feeds.

The data is cleaned, normalized, and stored in a database (MySQL) in a time-series format for further processing.

AI & Forecasting Module:

This module applies machine learning classifiers (Random Forest, XGBoost) for outbreak detection and deep learning models (LSTM/GRU) for forecasting disease trends 2–4 weeks ahead.

It also identifies anomalies using Isolation Forest and DBSCAN to flag unusual patterns early.

Composite Epidemic Risk Score (CERS) Module:

This module calculates a district-level risk score (0–100) by integrating climate data, population density, sanitation indicators, and historical disease trends.

The score provides a single, easy-to-interpret measure of epidemic risk for policymakers.

Explainable AI (XAI) Module:

SHAP and LIME are applied to ensure model transparency.

SHAP explains global feature importance, such as rainfall or temperature driving disease risk, while LIME gives local explanations for individual district forecasts.

Dashboard & GIS Visualization Module:

The system provides a web-based, interactive dashboard built in Streamlit.

GIS-enabled maps display district-level risk heatmaps, time-series forecasts, and CERS meters.

Users can visualize trends, compare districts, and explore detailed prediction insights.

Alert & Notification Module:

Threshold-based alerts are automatically generated when outbreak risks exceed defined levels.

Notifications are sent via email (SMTP) and SMS (Twilio API).

Weekly PDF reports summarizing high-risk districts, forecasted trends, and suggested preventive measures are also generated.

User Management & Authentication Module:

This module ensures secure access through login/logout functionality and role-based permissions.

Administrators, health officers, and general users have access only to the modules and data relevant to their role

3.2 Behaviour Requirements

Behavior requirements describe the behavior of the system between inputs and outputs.

This figure illustrates how the user interacts with the system for performing different functionalities. The behavioral requirements of our system as shown below.

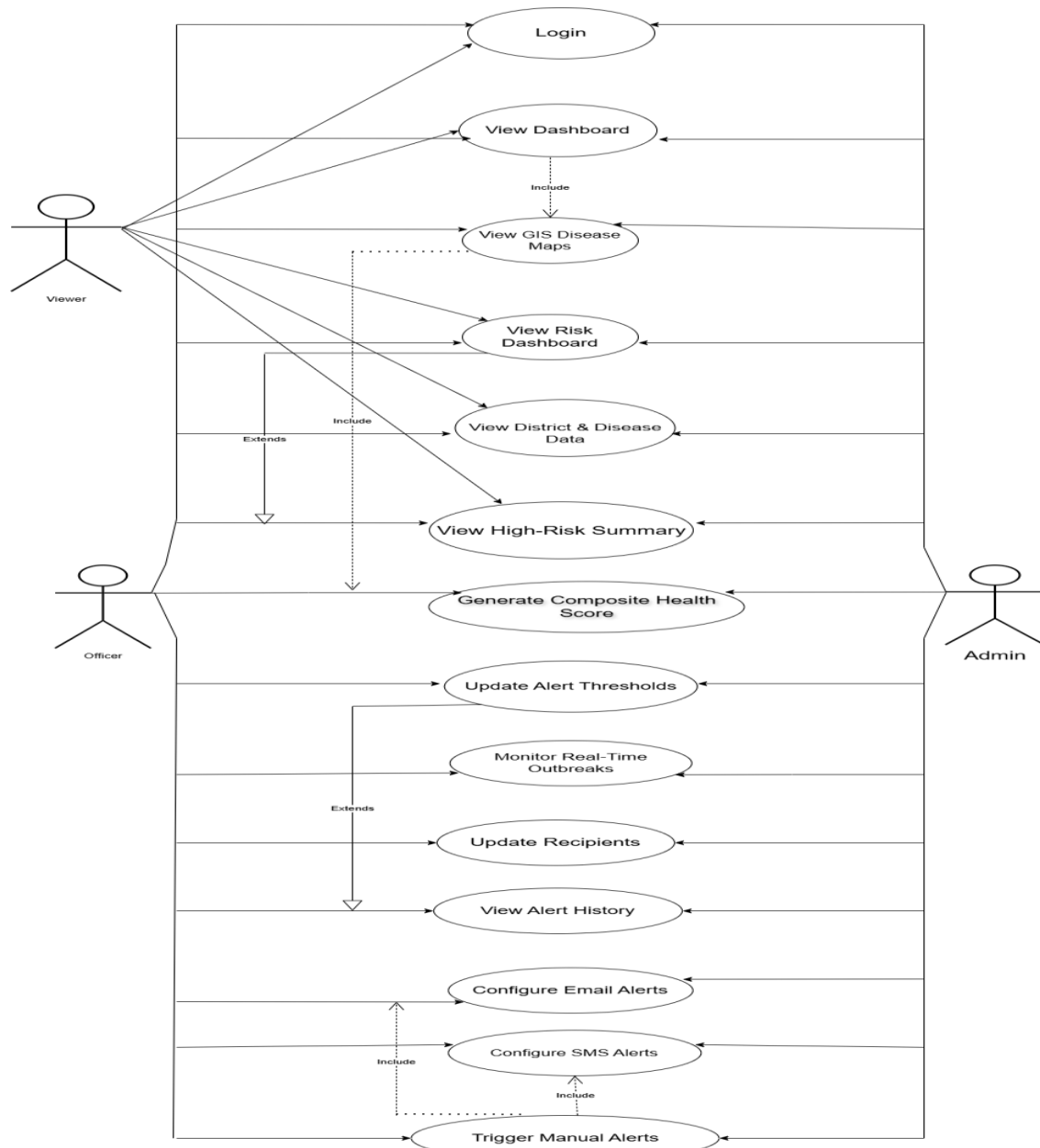


Figure 3.1 Usecase Diagram of PakPulse-AI

3.3 External Interface Requirements

No external interface

3.4 User Interfaces

Dashboard Screen

After login users are directed to their dashboard. Here, they can view different information

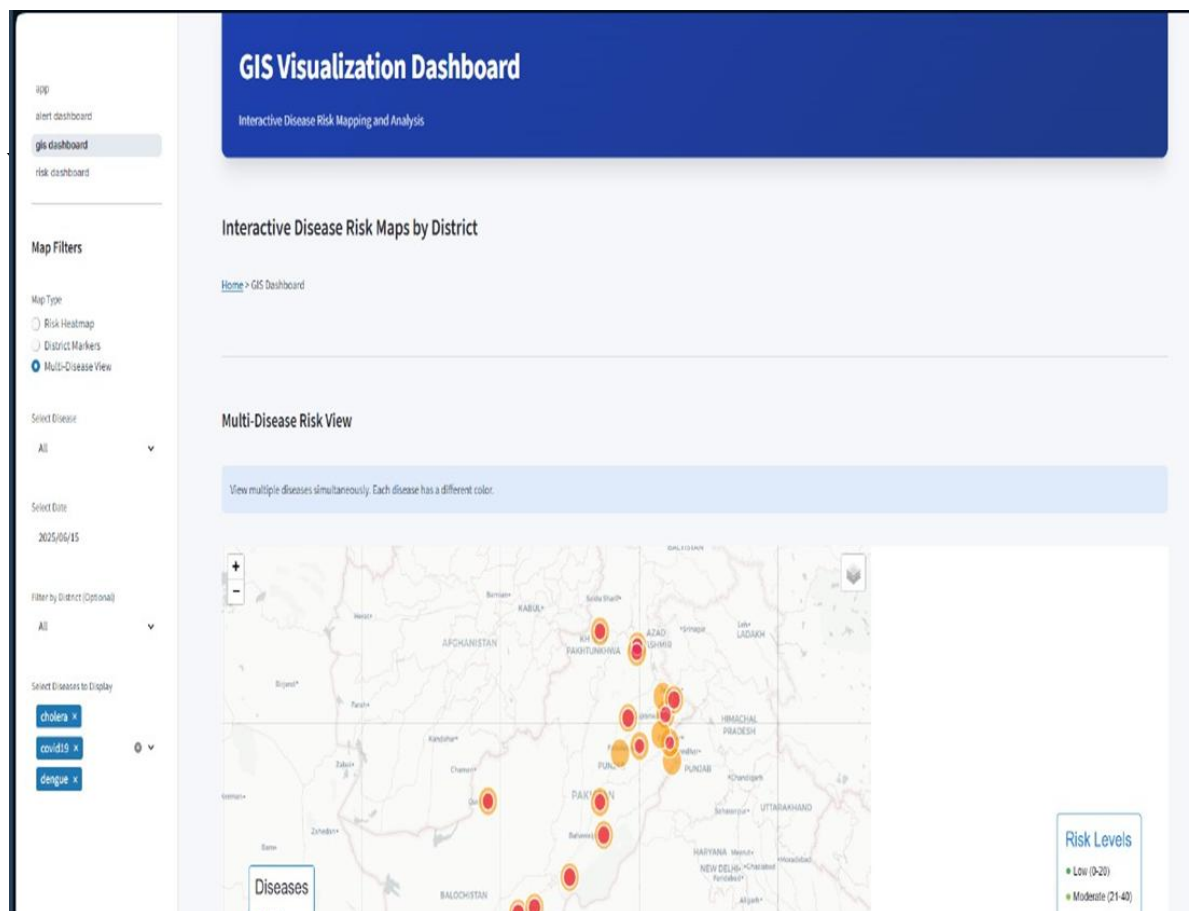


Figure 3.2 GIS Visuals

Admin Dashboard Screen

When admin login, a dashboard open up and show different features. Through these admin can perform various action and gain insights

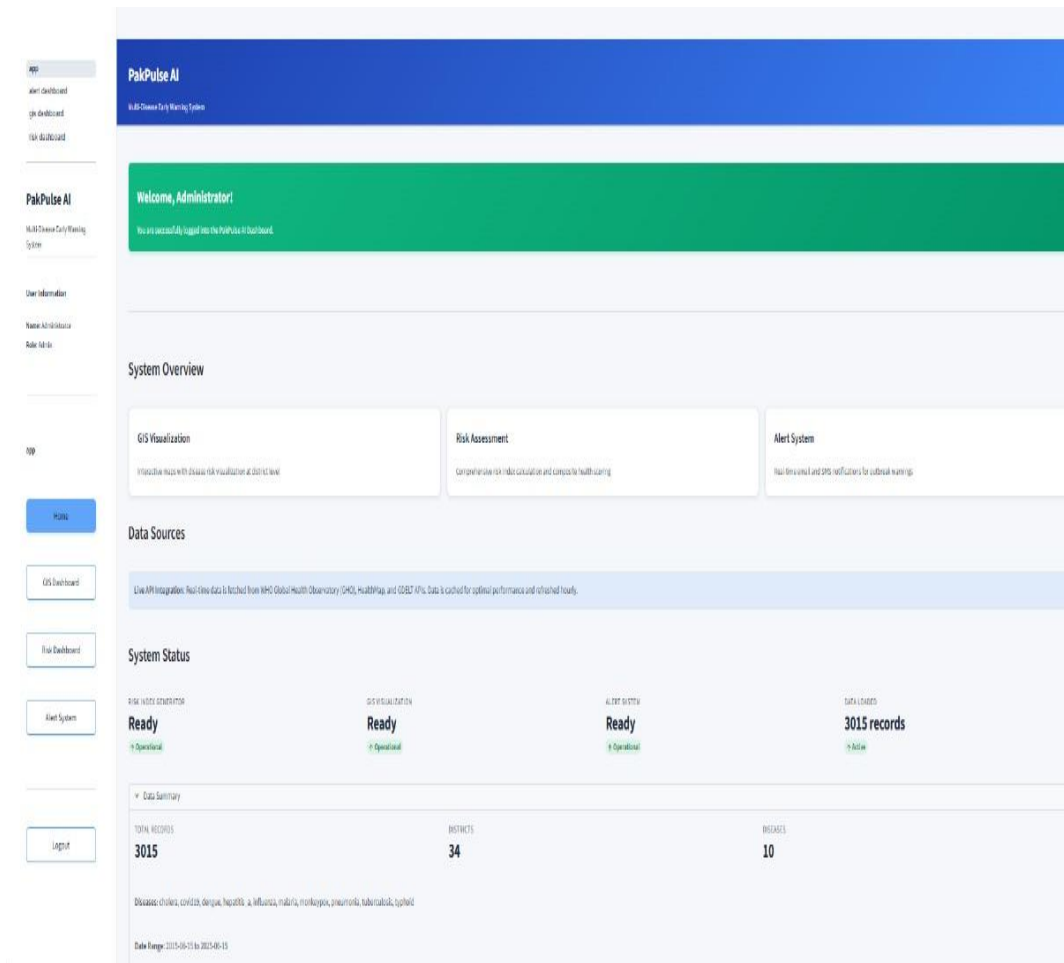


Figure 3.3 Dashboard

3.4.1 Hardware Interfaces

No hardware interfaces will be used in our system.

3.4.2 Software Interfaces

Following are the software interfaces which are used in our system.

Table 1 Software Interface		
Tools	Version	Relation
<i>VS Code</i>	<i>latest</i>	<i>Development Environment</i>
<i>API Endpoints</i>	<i>REST VI</i>	<i>Communication with external data sources</i>
<i>Internet</i>	<i>required</i>	<i>APIs and Cloud Communication</i>
<i>Email server</i>	<i>standard</i>	<i>Sent alert notification</i>
<i>Twilio SMS Gateway</i>	<i>latest</i>	<i>Sent SMS alerts</i>

3.4.3 Communications Interfaces

PakPulseAI communicates through several essential interfaces to ensure seamless data flow and real-time operations. The system uses RESTful APIs over HTTPS to fetch live disease, outbreak, and weather data from external sources such as WHO, NIH, GDELT, and HealthMap. It relies on SMTP (TLS/SSL) for sending automated email alerts to registered recipients and uses the Twilio SMS API over secure HTTPS to deliver critical risk notifications via SMS. All external data exchange and alert transmissions operate over standard TCP/IP internet communication protocols, while the interaction between the Streamlit-based frontend and backend services uses HTTP and WebSocket communication for real-time dashboard updates and user actions.

4 Other Non-functional Requirements

4.1 Performance Requirements

This section outlines the expected performance criteria for PakPulse AI, including response times, throughput, and resource utilization.

4.1.1 Response Time

The system shall not take more than 5 seconds to process the input requested from the user.

4.1.2 Availability

The system shall be operational 24/7 and maintain at least 98% uptime, allowing for up to 2% downtime during scheduled maintenance, typically on weekends.

4.1.3 Accuracy

The system shall provide accurate outbreak forecasts, risk scores, and alerts based on validated data. All modules must undergo extensive testing to ensure minimal bugs and correct functionality.

4.2 Safety and Security Requirements

The system shall ensure secure access by authenticating all users and preventing unauthorized login attempts. Role-based permissions will limit access to sensitive data and modules according to user type (Administrator, Health Officer/Analyst, General User).

4.3 Software Quality Attributes

4.3.1 Usability

The system shall provide an intuitive interface and straightforward navigation to enable seamless interactions
for all users..

4.3.2 User Interface

The dashboard and visualizations shall be clear, appealing, and easy to interpret.

4.3.3 Ease of Use:

Users should be able to navigate the system and access required features within 5 minutes of first use

4.3.4 Learnability

Users with basic computer skills should be able to efficiently use the system after 20 minutes of initial exploration, with less than 0.5% errors in task execution.

4.3.5 Reliability

PakPulse AI shall implement robust error-handling mechanisms and conduct regular maintenance to ensure consistent performance.

4.3.6 Availability

The system shall maintain 98% uptime, with up to 2% downtime for maintenance.

4.3.7 Robustness

In the event of a system failure, recovery time shall not exceed 10 minutes in 99% of cases.

4.3.8 Portability

The system should function across different devices and screen sizes, rendering all components accurately and maintaining full functionality whether accessed from desktops, laptops, or tablets

5 Design Description

5.1 Composite Viewpoint

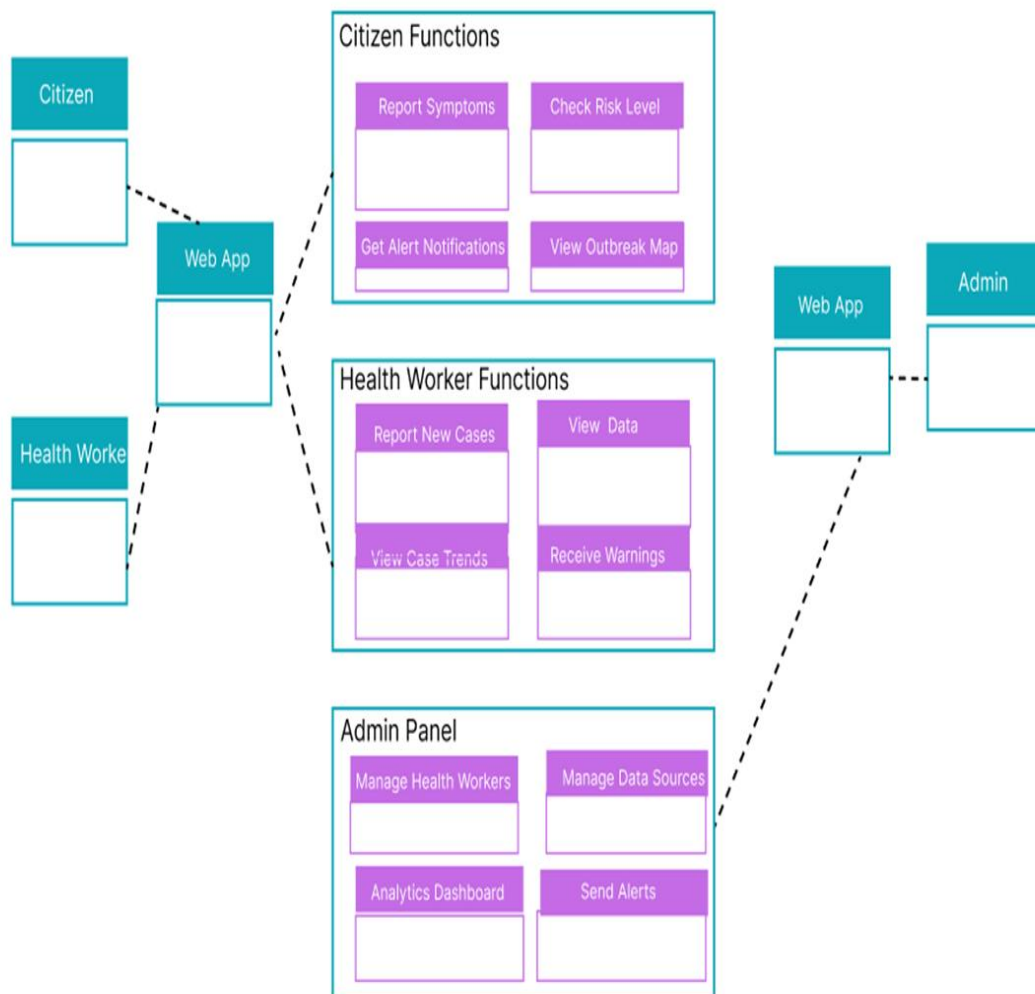


Figure 5.1 Composite view of PakPulse-AI

5.2 Logical Viewpoint

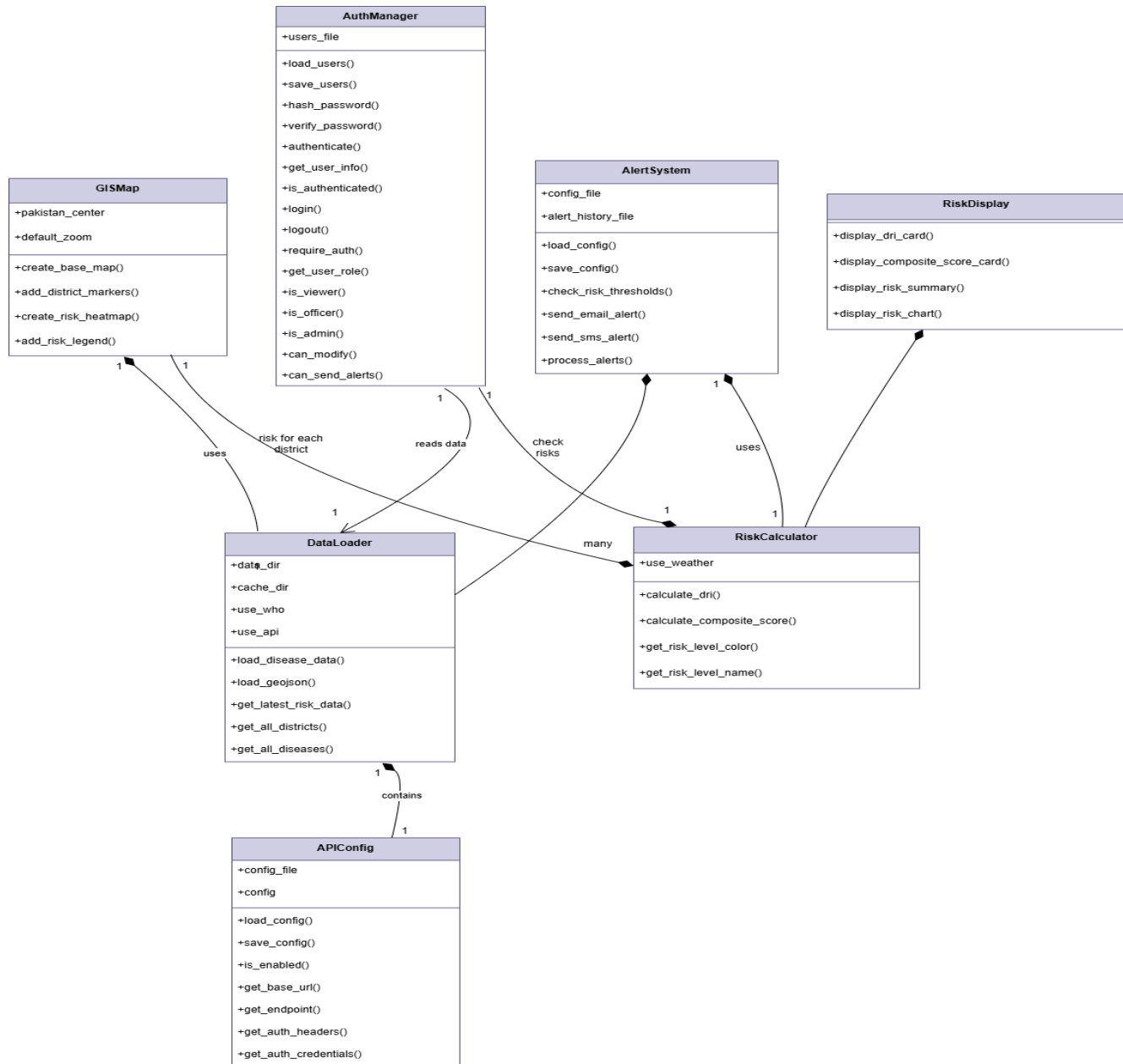


Figure 5.2 Class Diagram

5.3 Information Viewpoint

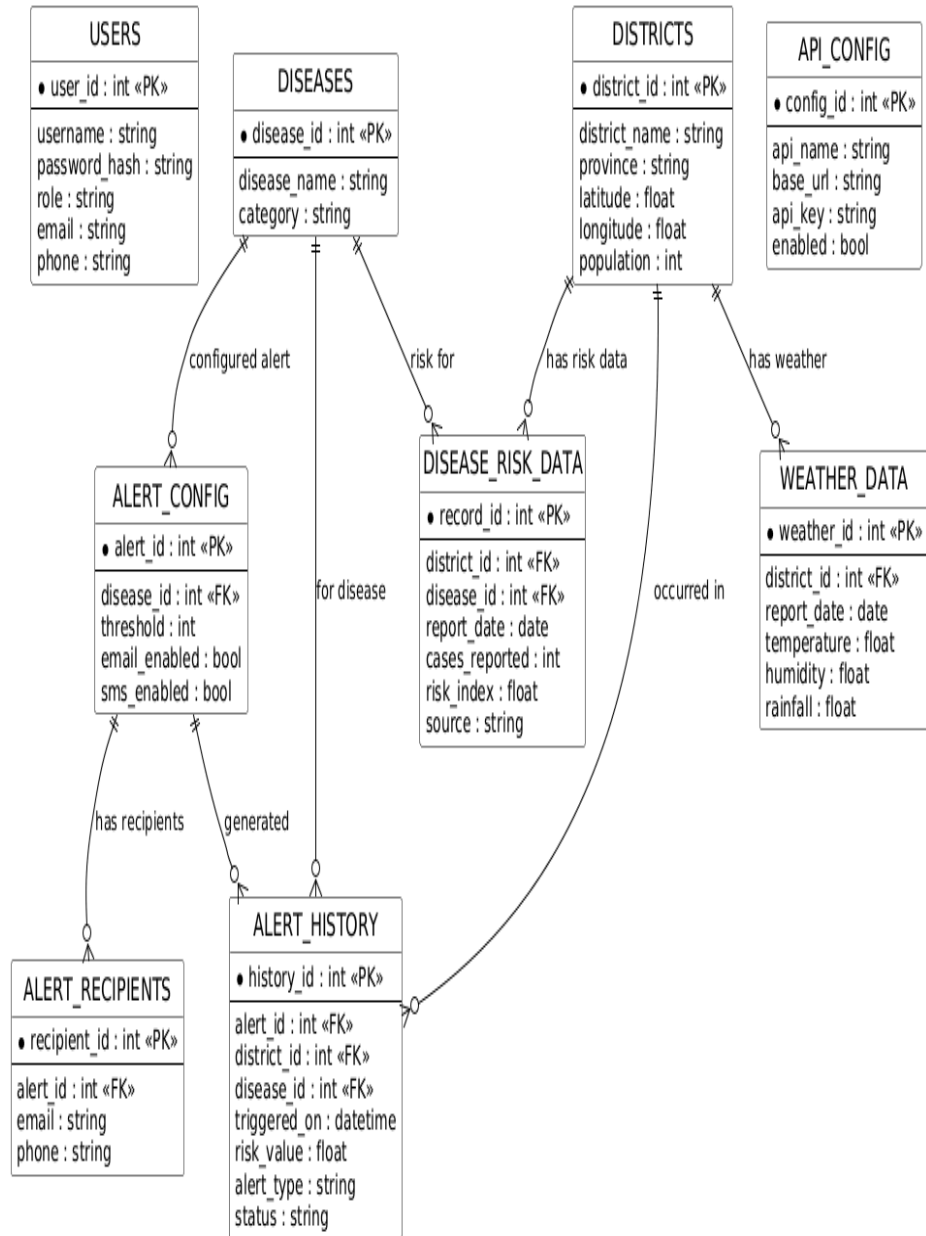


Figure 5.3 ER Diagram PakPulse-AI

5.4 Interaction Viewpoint

5.4.1 Sequence Diagram for General User

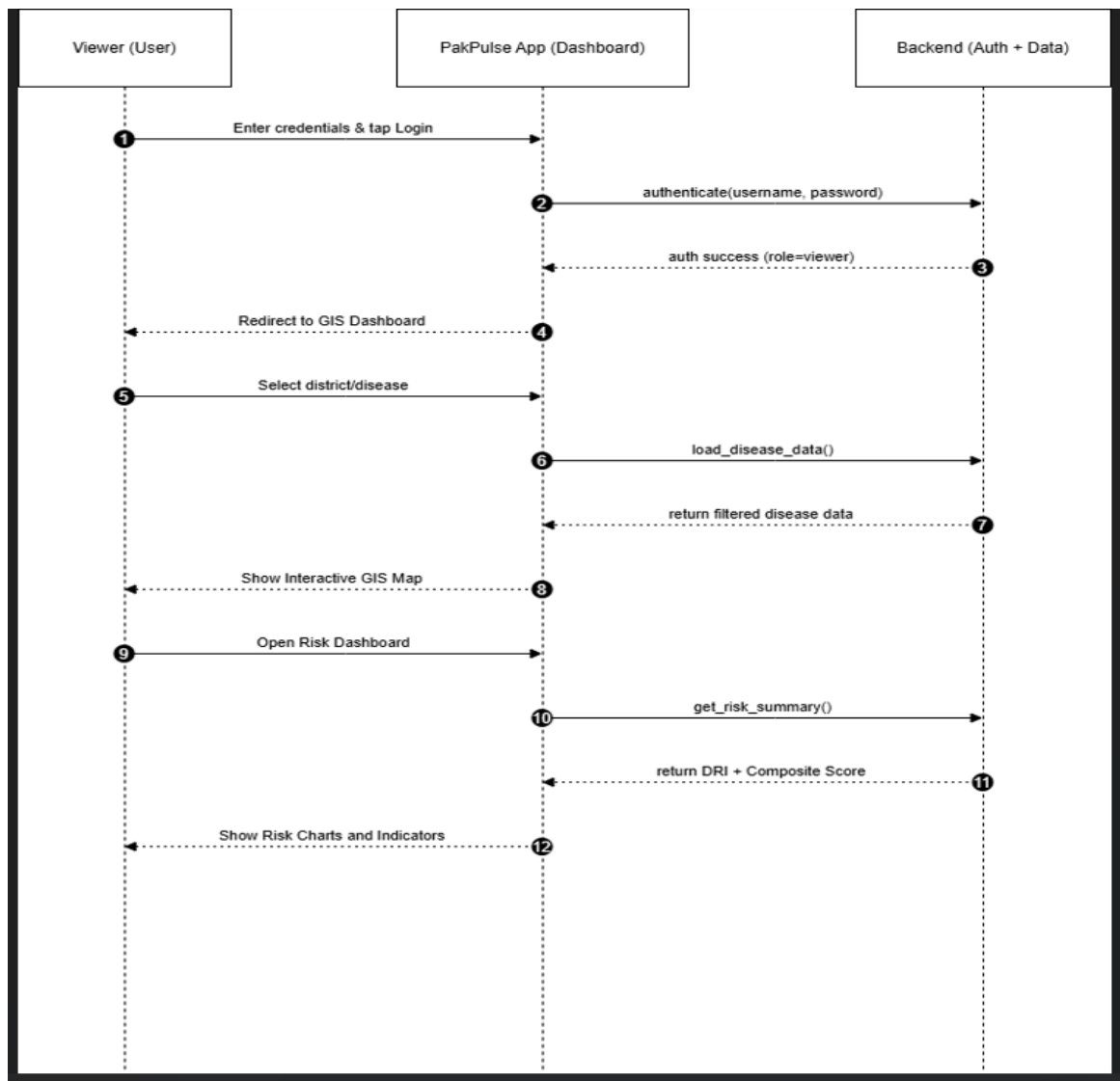


Figure 5.4 Sequence Diagram General User

5.4.2 Sequence Diagram for Admin

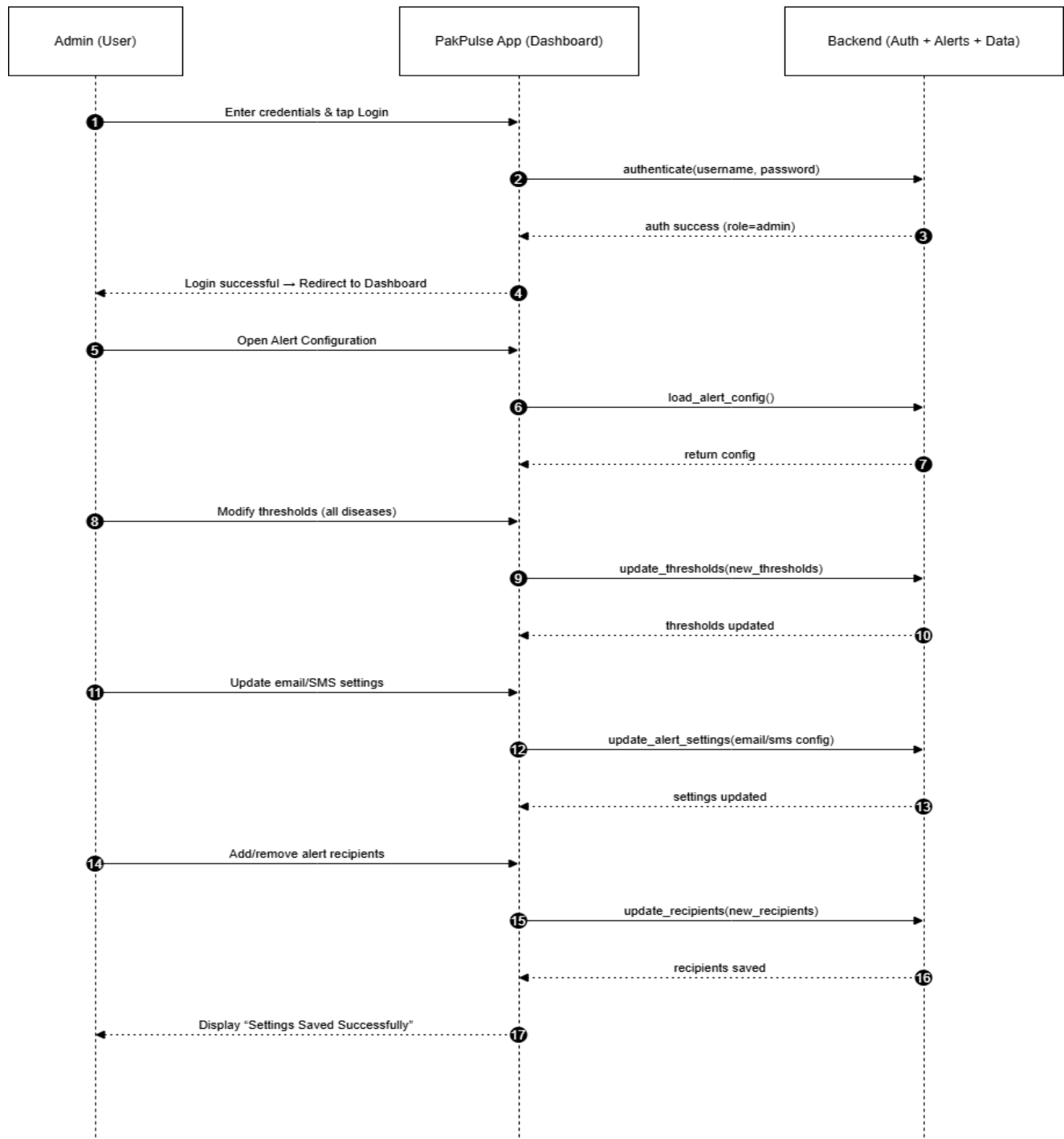


Figure 5.5 Sequence Diagram Admin

5.4.3 Sequence Diagram for Health Officer

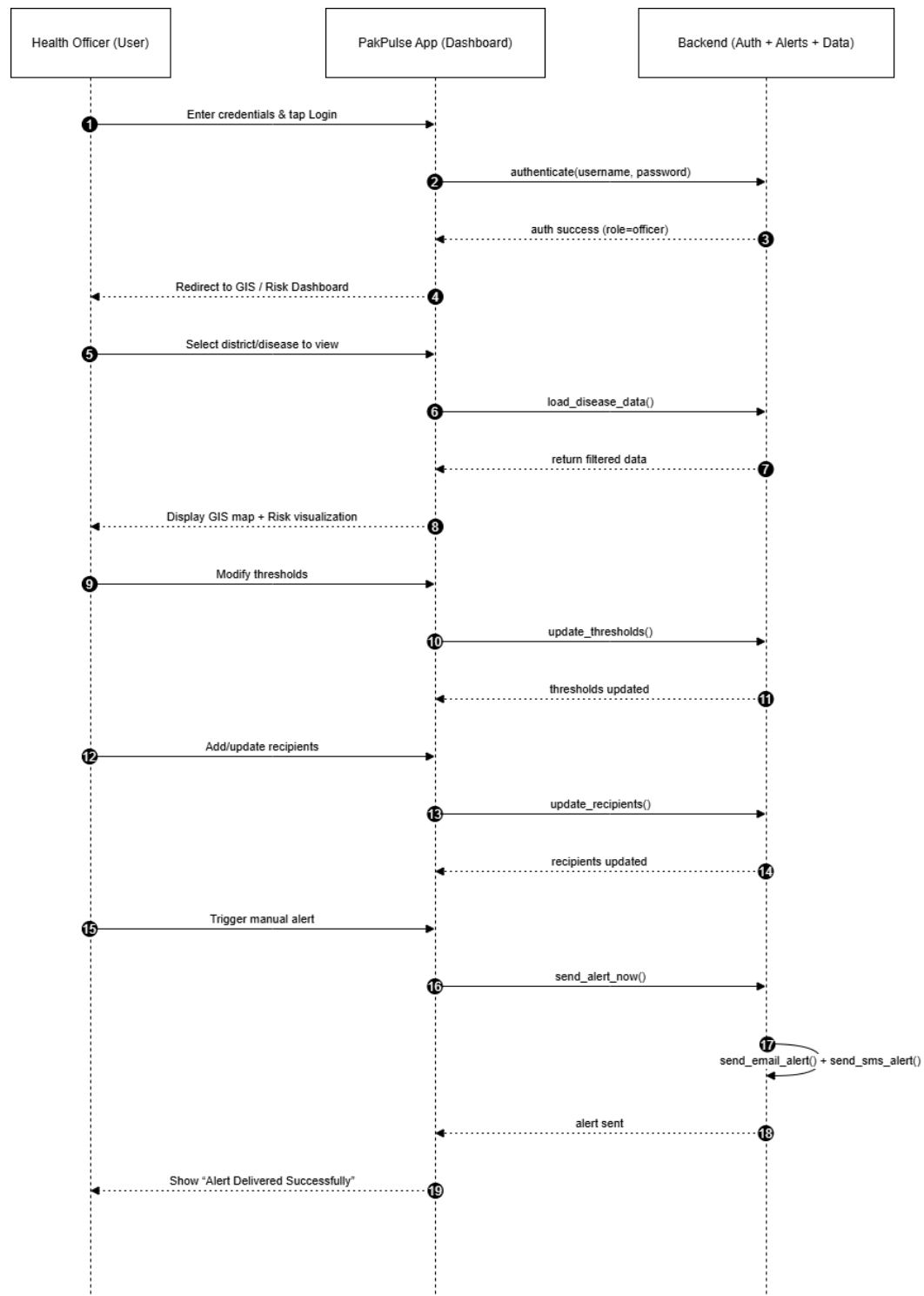


Figure 5.6 Sequence Diagram-Health Officer

5.5 State Dynamics Viewpoint

5.5.1 User State Diagram

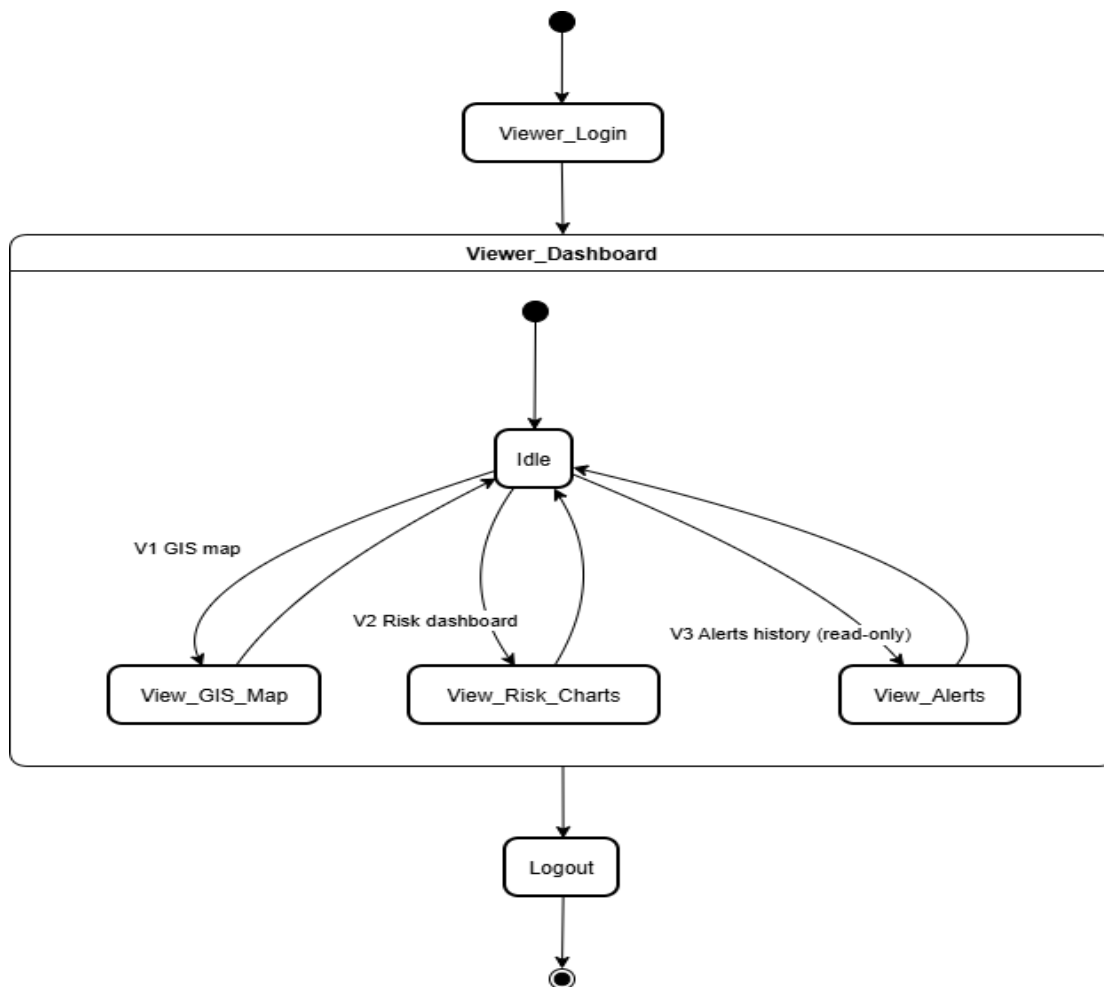


Figure 5.7 User State Diagram

5.5.2 Admin State Diagram

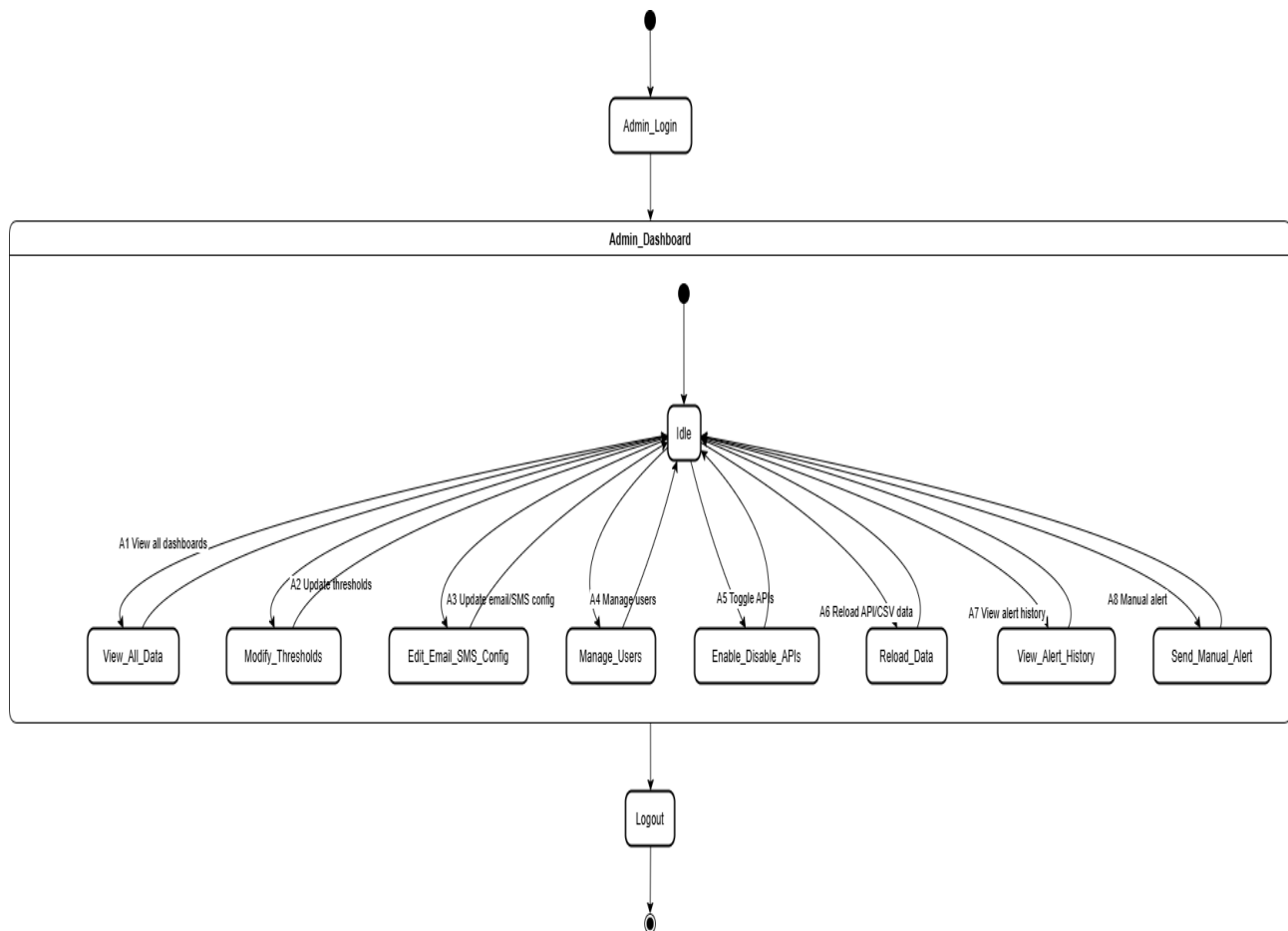


Figure 5.9 State Diagram Admin

5.5.3 Health Officer State Diagram

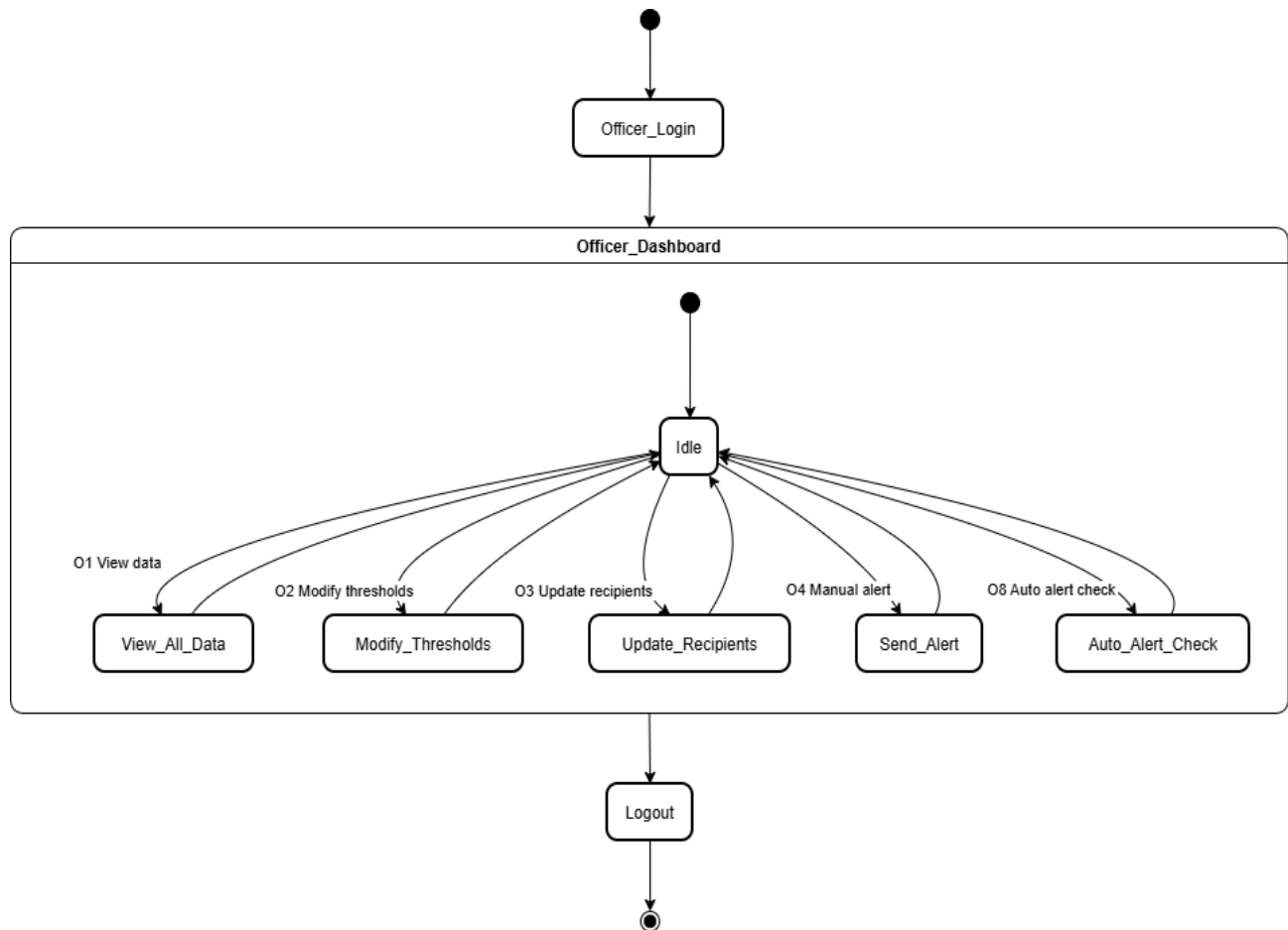


Figure 5.10 State Diagram Health Officer

5.6 Algorithm Viewpoint

5.6.1 Algorithmic Viewpoint — Login (General System Login):

Attributes	**Description**
-----	-----
Email / ID	Unique identifier for any user.
Password	Stored in hashed form.

Methods	**Description**
-----	-----
authenticate()	Checks credentials and returns corresponding user role.
redirectUser()	Sends the authenticated user to the correct dashboard.

Pseudo Code:

```

START
DEFINE function handleLogin(identifier, password)
    DISPLAY loading indicator

    FETCH user FROM UserDatabase WHERE id_or_email = identifier

    IF user DOES NOT EXIST
        DISPLAY "Invalid Credentials"
        RETURN
    ENDIF

    IF user.password MATCHES password
        DETERMINE user.role

        IF role == "Admin"
            NAVIGATE to AdminDashboard
        ELSE IF role == "HealthOfficer"
            NAVIGATE to OfficerDashboard
        ELSE IF role == "GeneralUser"
            NAVIGATE to UserDashboard
        ENDIF

        DISPLAY "Successfully Logged In"
    ELSE
        DISPLAY "Invalid Credentials"
    ENDIF
END

ON loginButton CLICK
    GET identifier from input
    GET password from input
    CALL handleLogin(identifier, password)

```

END

5.6.2 Algorithmic Viewpoint — Admin

Attributes	**Description**
Admin ID	Unique ID of the admin.
Name	Admin's full name.
Role	Always set to "Admin".
Permissions	Full system access.

Methods	**Description**
manageUsers()	Add, remove, or update system users.
updateModels()	Upload and configure predictive disease models.
setThresholds()	Adjust forecasting thresholds for alerts.
viewDashboard()	Access administrative dashboard and analytics.

Pseudo Code:

START

DEFINE function adminMenu(adminID)

DISPLAY "Admin Dashboard"

LOOP

DISPLAY options:

- 1 - Manage Users
- 2 - Update Prediction Models
- 3 - Configure Alert Thresholds
- 4 - View System Dashboard
- 5 - Logout

GET choice

IF choice == 1

CALL manageUsers()

ELSE IF choice == 2

CALL updateModels()

ELSE IF choice == 3

CALL setThresholds()

ELSE IF choice == 4

CALL viewDashboard()

ELSE IF choice == 5

EXIT LOOP

ENDIF

END LOOP

END

5.6.3 Algorithmic Viewpoint — Health Officer:

Attributes	**Description**
Officer ID	Unique identifier for the health officer.
Region Assigned	Officer's jurisdiction.
Role	"HealthOfficer/Analyst".

Methods	**Description**
uploadData()	Submit new case data or surveillance datasets.
analyzeTrends()	View outbreak patterns and graphs.
generateReports()	Produce situation reports and summaries.
viewForecasts()	Check predicted disease cases for coming weeks.

Pseudo Code:

START

DEFINE function officerMenu(officerID)

DISPLAY "Health Officer Dashboard"

LOOP

DISPLAY options:

- 1 - Upload Disease Data
- 2 - Analyze Trends
- 3 - Generate Reports
- 4 - View Forecasts
- 5 - Logout

GET choice

IF choice == 1

CALL uploadData()

ELSE IF choice == 2

CALL analyzeTrends()

ELSE IF choice == 3

CALL generateReports()

ELSE IF choice == 4

CALL viewForecasts()

ELSE IF choice == 5

EXIT LOOP

ENDIF

END LOOP

END